

BUSINESS REQUIREMENTS DOCUMENT (BRD)

Project Title:

Supply Chain Performance & Risk Analysis Dashboard for DataCo Global

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Version: 1.0

1. Project Objective

The purpose of this project is to explore the DataCo Supply Chain dataset, perform detailed data assessment and cleaning, build a robust data model in Power BI, and design an interactive dashboard. The dashboard will provide end-to-end visibility into supply chain operations, highlight bottlenecks, and deliver actionable insights to improve On-Time Delivery (OTD), reduce shipment delays, control logistics costs, and increase profitability.

2. Business Problem / Opportunity

DataCo Global is facing several critical supply chain challenges:

- Frequent shipment delays causing stockouts and lost sales
- Rising logistics and shipping costs
- Inconsistent reliability across shipping modes and regions
- High late delivery risk affecting customer satisfaction
- Lack of a single unified view to monitor overall supply chain health

This project aims to address these issues by transforming raw transactional data into meaningful insights and visualizations for better decision-making.

3. Target Audience

- Supply Chain Managers and Operations Teams
- Logistics and Shipping Teams
- Senior Management and Executives
- Procurement and Planning Teams
- Business Analysts

4. Scope

In Scope:

- Loading and detailed assessment of the DataCoSupplyChainDataset.csv
- Data cleaning and transformation using Power Query
- Creation of calculated columns and star schema data model in Power BI
- Development of an interactive multi-page dashboard
- Analysis of On-Time Delivery, delays, profitability, shipping efficiency, and regional performance

Out of Scope:

- Predictive modeling and forecasting
- Real-time data integration
- PostgreSQL database design and normalization
- Mobile application development

5. Data Source(s)

Dataset Name: DataCoSupplyChainDataset.csv

Source: Kaggle – DataCo SMART SUPPLY CHAIN FOR BIG DATA ANALYSIS

Size: Approximately 180,519 rows and 53 columns (~95 MB)

Type: Structured transactional CSV file containing orders, shipping, delivery status, product categories, customer segments, profit, and late delivery risk

6. Key Metrics / KPIs

The dashboard will track the following key metrics:

- On-Time Delivery (OTD) Percentage
- Late Delivery Rate and Average Delay Days
- Total Sales and Total Profit
- Profit Margin %
- Shipping Performance by Mode and Region
- Late Delivery Risk by Product Category, Customer Segment, and Order Region
- Monthly, Quarterly, and Year-over-Year trends in sales, profit, and delivery performance

7. Deliverables

- BRD (Business Requirements Document)
- Dataset Assessment Report and Cleaned Dataset
- Functional Requirements Document (FRD) and Dashboard Mockups
- Power BI Dashboard (.pbix file) with proper data model
- Final Analysis Report with insights and recommendations
- README documentation

8. Project Timeline

- **Day 1:** Dataset Understanding and BRD Submission
- **Day 2:** Data Cleaning and Preparation
- **Day 3:** Power BI Connection, Data Model, and DAX Measures
- **Day 4:** Dashboard Design and Development
- **Day 5:** Finalization, Testing, and Reporting